



A 3D thermal LB model on non-orthogonal grid and its application for natural convection in irregular domains

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ABSTRACT

In this paper, a 3D thermal LB model on non-orthogonal grid is derived, as well as the corresponding boundary treatment method for irregular boundary. Numerical validation is then performed by modeling natural convection in a 3D annulus. It is proved that the thermal flow structure in the annulus is accurately predicted with less grids. After that, present method is applied to study natural convection in a 3D cavity with inclined side walls. Two opposite inclined walls on the left and right sides are kept at different constant temperatures, while the remained four walls of the cavity are either adiabatic (Type 1) or own linear temperature variations (Type 2). It is found out that natural convection in the inclined cavity exhibits distinct 3D features under Type 2 thermal boundary setting, yet it can be viewed as a 2D process under Type 1 condition. Moreover, a large Prandtl number ($Pr = 10.0$) can generally suppress the 3D effect stemmed from thermal boundary setting. The aforementioned results indicate that present model and boundary treatment method are reliable to deal with 3D natural convection in irregular domains.

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1. Introduction

As a novel computational fluid dynamics (CFD) approach, the lattice Boltzmann method (LBM) has attracted increasingly attention in the past few decades. Different from the conventional CFD methods, the LBM is based on the discrete Boltzmann equation in statistical physics and it is generally represented by means of a sequence of collide-and-stream steps. Such a kinematic nature of the LBM makes it easier to incorporate microscopic characteristics [1]. As a minimal set of discrete velocities is used in the LBM, the transformation between microscopic particle distribution function and macroscopic quantities is greatly simplified [2–4]. Hence, the LBM has higher computing efficiency compared with the particle-based numerical methods, such as Molecular dynamics (MD) and direct simulation of Monte Carlo (DSMC) [5]. So far, the LBM has been established to be an effective and powerful tool for modeling single/multi-phase fluid flow and heat transfer in various hydrodynamic systems [1,6–10].

Inheriting from the Lattice Gas Automaton (LGA) [11], a symmetrical and uniform grid is required in the standard LB model to ensure that particles can arrive at their adjacent nodes over a

time step. However, for the research domain with irregular geometry, the uniform grid with stair-shaped approximation has difficulty in implementing the boundary condition accurately and effectively. To eliminate this shortcoming, accurate boundary treatment methods, including adaptive meshes refinement method and inter/extrapolated curved boundary method, are proposed to deal with the curved boundaries [12–15]. It should be pointed out that these boundary schemes, with different numerical stabilities, have their own validities [16]. Alternatively, it is a pretty good way to choose non-orthogonal grids, such as body-fitted grids, to discretize the irregular domain, since the grid nodes can be directly located on the boundaries. Meanwhile, great efforts have been made to develop the LBM strategies on non-orthogonal grid, which can be roughly classified into differential type and interpolation type. For the differential type, the LB equations are transformed into a set of differential equations, and then they are solved by conventional numerical method including finite difference method (FDM) [17,18], finite element method (FEM) [19,20], and finite volume method (FVM) [21]. As for the interpolation type, initial attempt was done in the work of He and Doolen [22], where a interpolation-supplemented lattice Boltzmann method (ISLBM) was proposed. The interpolation step was added to determine the post-streaming particle distribution functions at grid nodes. But the requirement of an analytical function to govern the transformation from physical plane to computational plane limits the

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broad application of the ISLBM. Later, by introducing the body-fitted coordinates into the LBM, Imamura et al. [23,24] proposed the generalized form of interpolation-supplemented lattice Boltzmann method (GILBM), which could be applied on non-orthogonal grid with no limitation. As the local time step is adopted, the GILBM owns a high computational efficiency. Similarly, by using the complete transformation approach, Budinski [25] derived a MRT-LBM model for 2D flows in curvilinear coordinates. The model was successfully applied on physical and schematic domains of complex shape. The other LBM scheme for generalized coordinates and curved spaces was presented in the work of Mendoza et al. [26]. The discrete Boltzmann equation was solved in contravariant coordinates, and the equilibrium distribution function was obtained by a Hermite polynomials expansion.

More recently, Li et al. [27] assessed the GILBM thoroughly by studying the fluid flow in irregular domains. It was found out that as the Reynolds number increases, both the accuracy and efficiency of the GILBM have a better performance than the standard LBM. Moreover, the GILBM can easily incorporate with the local grid refinement technique by using a coarse grid in whole domain and a finer one in critical region, thus tuning the balance between computing efficiency and accuracy [28]. Consequently, a GILBM-based LB model was then proposed in the other work of Li et al. [29]. It was successfully utilized to simulate 2D natural and forced convective heat transfer in irregular domains showing great numerical stability under high Rayleigh number.

Note that most of the above-mentioned studies focus on the heat and fluid flow in 2D irregular domains. It is true that the 2D simulations are adoptable for the cases in which the effect of the third dimension can be ignored. However, under some circumstances, for example, when the radius ratio (RR) of an inner heated 3D annulus is limited within the range of $1.24 \leq RR \leq 1.71$, counter-rotating transverse rolls will occur in its upper region along the direction of the axis, for which the dimensional effect

rectangular cavity [31–33]. Besides, Karimipour et al. [34,35] investigated the effects of inclination angle, Prandtl Number, and boundary condition on the mixed convection in the rectangular cavity. Nevertheless, the natural convection in a 3D cavity with inclined side walls is rarely involved in the previous investigations. Hence, it is considered in the current work, for which the effects of boundary setting and Prandtl number are revealed to provide a valuable guidance for the practical application.

The rest of this paper is organized as follows: In Section 2, the 3D thermal LB model on non-orthogonal grid is derived in detail. In Section 3, the corresponding boundary treatment method is presented. In Section 4, numerical simulations and results are shown and discussed. Finally, some conclusions are summarized in Section 5.

2. Methodology

2.1. 3D thermal LB model on uniform Cartesian grid

Two sets of particle distribution functions, f_i and h_i , are used to characterize the velocity field and temperature field in the 3D thermal LB model on uniform Cartesian grid, respectively. The evolution equations are given as

$$f_i(\mathbf{x} + \mathbf{e}_i \delta t, t + \delta t) - f_i(\mathbf{x}, t) = -\frac{1}{\tau_f} [f_i(\mathbf{x}, t) - f_i^{eq}(\mathbf{x}, t)] + \delta t F_i \quad (1)$$

$$h_i(\mathbf{x} + \mathbf{e}_i \delta t, t + \delta t) - h_i(\mathbf{x}, t) = -\frac{1}{\tau_h} [h_i(\mathbf{x}, t) - h_i^{eq}(\mathbf{x}, t)] \quad (2)$$

where $\mathbf{x}$ is the lattice position, t is the time, and δt is the time step. τ_f and τ_h stand for the dimensionless relaxation times of f_i and h_i , respectively. As the three-dimensional nineteen-velocity (D3Q19) model [36] is chosen, the discrete velocity vectors $\mathbf{e}_i = (e_{i,1}, e_{i,2}, e_{i,3})$ shown in Fig. 1 are set as

$$\mathbf{e}_i = c \begin{bmatrix} 0 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 & 0 & 0 & 1 & -1 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 & -1 & 1 & -1 & 0 & 0 & -1 & 1 & -1 & 1 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 & -1 & 0 & 0 & 1 & -1 & -1 & 1 & 0 & 0 & 1 & -1 \end{bmatrix} \quad (3)$$

must be considered [30]. Therefore, it is necessary to extend the GILBM to 3D application given its distinct advantages in dealing with irregular boundary and improving computing accuracy. Unfortunately, to the best of our knowledge, few works have been reported in this aspect. The purpose of current paper is to develop a 3D thermal LB model on non-orthogonal grid based on the GILBM. In order to achieve this goal, the corresponding Jacobian conversion for the 3D GILBM-based thermal model should be reconstructed. In addition, assisted with a group of transformation equations deduced in this paper, a boundary treatment method on non-orthogonal grid is obtained to make present model cover more types of boundary conditions. After that, the developed model and boundary treatment method are applied to modeling natural convection in 3D enclosed structures, for its important role playing in many engineering applications, such as cooling of electronic devices and energy storage systems [30]. Concretely, natural convection in a 3D horizontal concentric air-filled annulus is simulated first for numerical validation. Then, in terms of the natural convection in the cavities, most of the studies focus on the

where $c = \delta x / \delta t$ is the lattice speed and it is related to the sound speed c_s with $c_s^2 = c^2 / 3$.

The corresponding equilibrium distribution functions f_i^{eq} and h_i^{eq} are defined as

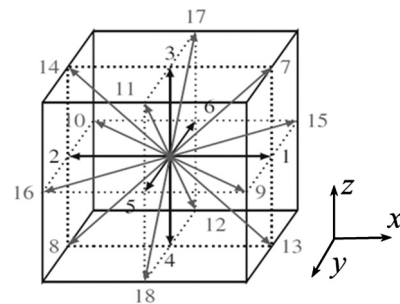


Fig. 1. Velocity set of D3Q19 model.

$$f_i^{eq} = \rho \omega_i \left[1 + \frac{\mathbf{e}_i \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_i \cdot \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u}^2}{2c_s^2} \right] \quad (4)$$

$$h_i^{eq} = T \omega_i \left[1 + \frac{\mathbf{e}_i \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_i \cdot \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u}^2}{2c_s^2} \right] \quad (5)$$

where ρ , $\mathbf{u}$, and T are the macroscopic density, velocity, and temperature, respectively. ω_i is the weight coefficient with the value set as

$$\omega_i = \begin{cases} 1/3, & i = 0 \\ 1/18, & i = 1, 2, \dots, 6 \\ 1/36, & i = 7, 8, \dots, 18 \end{cases} \quad (6)$$

The force term F_i in Eq. (1) is added to couple the velocity field with the temperature field [37], and it is given as

$$F_i = \mathbf{G} \cdot \frac{(\mathbf{e}_i - \mathbf{u})}{p} f_i^{eq} \quad (7)$$

with $\mathbf{G} = -\beta(T - T_0)\mathbf{g}$. Here, β is the expansion coefficient, T_0 is the reference temperature, $\mathbf{g}$ is the acceleration caused by gravity. By using such a force term, the gravity effect is considered with no other modifications [38,39].

Macroscopic parameters are calculated by

$$\rho = \sum_i f_i, \quad \rho \mathbf{u} = \sum_i f_i \mathbf{e}_i, \quad T = \sum_i h_i \quad (8)$$

The dimensionless relaxation times, τ_f and τ_T are related to the kinematic coefficient of viscosity ν and thermal diffusion coefficient χ as

$$\nu = c_s^2 \left(\tau_f - \frac{1}{2} \right) \delta t, \quad \chi = c_s^2 \left(\tau_T - \frac{1}{2} \right) \delta t \quad (9)$$

2.2. 3D thermal LB model on non-orthogonal grid

Based on the GILBM, the above-mentioned 3D thermal LB model on uniform Cartesian grid is extended on non-orthogonal grid. As shown in Fig. 2, the governing equations in the physical domain $\mathbf{x} = (x_1, x_2, x_3) = (x, y, z)$ should be transformed to the computational domain $\xi = (\xi_1, \xi_2, \xi_3) = (\xi, \eta, \gamma)$. Firstly, Eq. (1) is divided into

$$\text{Collision: } f_i^*(\mathbf{x}, t) = -\frac{1}{\tau_f} [f_i(\mathbf{x}, t) - f_i^{eq}(\mathbf{x}, t)] + \delta t F_i \quad (10)$$

$$\text{Streaming: } f_i(\mathbf{x}, t + \delta t) = f_i^*(\mathbf{x} - \mathbf{e}_i \delta t, t) \quad (11)$$

where f_i^* is the post-collision particle distribution function of density. As only local particle distribution functions involved in the collision step, the transformation of collision equation is easily achieved by replacing $\mathbf{x}$ with ξ as follows

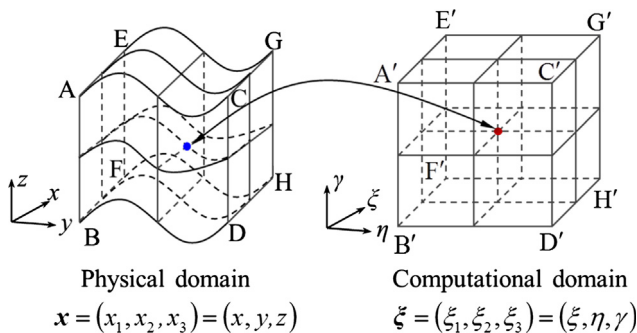


Fig. 2. Relation between physical and computational domains.

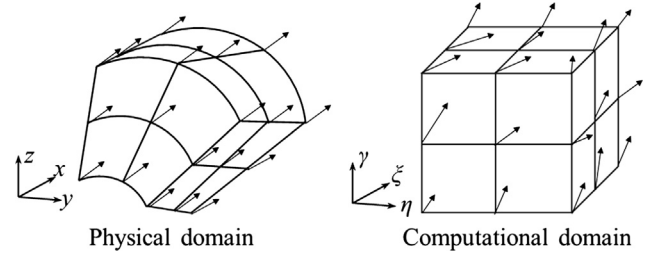


Fig. 3. Discrete velocity vectors in the physical and computational domains.

$$f_i^*(\xi, t) = -\frac{1}{\tau_f} [f_i(\xi, t) - f_i^{eq}(\xi, t)] + \delta t F_i \quad (12)$$

Fig. 3 shows the distribution of discrete velocity vector $\mathbf{e}_7 = c(1, 0, 1)$ in the physical and computational domains, respectively. Compared with the constant one in the physical domain, the discrete velocity vector varies with grid node in the computational domain. Hence, the discrete velocity in the computational domain is called contravariant velocity $\tilde{\mathbf{e}}_i = (\tilde{e}_{i,1}, \tilde{e}_{i,2}, \tilde{e}_{i,3})$ with the definition as

$$\tilde{e}_{i,\alpha} = \mathbf{e}_{i,\beta} \frac{\partial \xi_\alpha}{\partial x_\beta}, \quad (\alpha, \beta = 1, 2, 3) \quad (13)$$

The derivatives in the right-hand side of Eq. (13) are calculated by a developed Jacobian conversion as

$$\frac{\partial \xi_\alpha}{\partial x_\beta} = \frac{A_{\alpha\beta}}{J} \quad (14)$$

where J is the Jacobian determinant, $A_{\alpha\beta}$ is a transformation matrix, and they are defined as

$$J = \begin{vmatrix} x_\xi & y_\xi & z_\xi \\ x_\eta & y_\eta & z_\eta \\ x_\gamma & y_\gamma & z_\gamma \end{vmatrix}, \quad A_{\alpha\beta} = \begin{bmatrix} y_\eta z_\gamma - y_\gamma z_\eta & x_\gamma z_\eta - x_\eta z_\gamma & x_\eta y_\gamma - x_\gamma y_\eta \\ y_\gamma z_\xi - y_\xi z_\gamma & x_\xi z_\gamma - x_\gamma z_\xi & x_\gamma y_\xi - x_\xi y_\gamma \\ y_\xi z_\eta - y_\eta z_\xi & x_\eta z_\xi - x_\xi z_\eta & x_\xi y_\eta - x_\eta y_\xi \end{bmatrix} \quad (15)$$

In order to maintain the accuracy in space, second-order central-difference scheme is used to calculate the derivatives in Eqs. (13)–(15). Subsequently, the particle streaming distance $\Delta \xi_{up,i}$ is obtained by integrating the contravariant velocity over a time step as

$$\Delta \xi_{up,i} = \int_0^{\delta t} d\xi_i = \int_0^{\delta t} \tilde{\mathbf{e}}_i dt \quad (16)$$

For suppressing the numerical errors to the same order of the truncation error of LBE, the two-step Runge-Kutta method [23] is applied to calculate the integration in Eq. (16) as

$$1^{\text{st}} \text{ step: } \Delta \xi_{up,i}^{(1)} = \frac{1}{2} \delta t \tilde{\mathbf{e}}_i(\xi) \quad (17a)$$

$$2^{\text{nd}} \text{ step: } \Delta \xi_{up,i} = \delta t \tilde{\mathbf{e}}_i(\xi - \Delta \xi_{up,i}^{(1)}) + O(\delta t^3) \quad (17b)$$

Despite that the contravariant velocity and its integration should be calculated in advance, their values do not change, and the calculation just needs to be done once. Then, the streaming equations in the computational domain become

$$f_i(\xi, t + \delta t) = f_i^*(\xi - \Delta \xi_{up,i}, t) \quad (18)$$

In Eq. (17b) and Eq. (18), generally, the positions of $\xi - \Delta \xi_{up,i}$ and $\xi - \Delta \xi_{up,i}^{(1)}$ that depend on the contravariant velocity are not exactly located at the lattice points, thus an interpolation procedure is extremely needed. As suggested by Imamura et al. [23], the

second-order upwind quadratic interpolation scheme is adopted to suppress numerical dissipation. So, the Eq. (18) is rewritten as

$$f_i(\xi, t + \delta t) = f_i^*(\xi - \Delta \xi_{up,i}, t) \\ = \sum_{k=0}^2 \sum_{l=0}^2 \sum_{s=0}^2 a_{i,k,\xi} a_{i,l,\eta} a_{i,s,\gamma} f_i^*(\xi_{m+k-md}, \eta_{n+l-nd}, \gamma_{p+s-pd}, t) \quad (19)$$

where $md = \text{sgn}(\Delta \xi_{up,i,\xi})$, $nd = \text{sgn}(\Delta \xi_{up,i,\eta})$, and $pd = \text{sgn}(\Delta \xi_{up,i,\gamma})$ are the coefficients to determine the grid nodes for interpolation. $a_{i,k,\xi}$, $a_{i,l,\eta}$, and $a_{i,s,\gamma}$ are the interpolation coefficients. For the internal grid nodes, they are given as

$$\begin{aligned} a_{i,0,\alpha} &= \frac{1}{2} (|\Delta \xi_{up,i,\alpha}| - 1) (|\Delta \xi_{up,i,\alpha}| - 2) \\ a_{i,1,\alpha} &= -|\Delta \xi_{up,i,\alpha}| (|\Delta \xi_{up,i,\alpha}| - 2) \\ a_{i,2,\alpha} &= \frac{1}{2} |\Delta \xi_{up,i,\alpha}| (|\Delta \xi_{up,i,\alpha}| - 1) \end{aligned} \quad (20)$$

In terms of the neighbor-boundary grid nodes, the interpolation coefficients are calculated by

$$\begin{aligned} a_{i,0,\alpha} &= \frac{1}{2} \Delta \xi_{up,i,\alpha} (\Delta \xi_{up,i,\alpha} - 1) \\ a_{i,1,\alpha} &= 1 - \Delta \xi_{up,i,\alpha}^2 \\ a_{i,2,\alpha} &= \frac{1}{2} \Delta \xi_{up,i,\alpha} (\Delta \xi_{up,i,\alpha} + 1) \end{aligned} \quad (21)$$

Similarly, we can acquire the governing equation of temperature in the computational domain as

$$h_i^*(\xi, t) = -\frac{1}{\tau_T} [h_i(\xi, t) - h_i^{eq}(\xi, t)] \quad (22)$$

$$h_i(\xi, t + \delta t) = h_i^*(\xi - \Delta \xi_{up,i}, t) \\ = \sum_{k=0}^2 \sum_{l=0}^2 \sum_{s=0}^2 a_{i,k,\xi} a_{i,l,\eta} a_{i,s,\gamma} h_i^*(\xi_{m+k-md}, \eta_{n+l-nd}, \gamma_{p+s-pd}, t) \quad (23)$$

where h_i^* is the post-collision particle distribution function of temperature.

As the grid size in the present model is not uniform, a local time step δt can be applied to accelerate the solution to the converged steady state, and it is determined by the Courant-Friedrichs-Lewy (CFL) condition [40] as

$$\delta t = \lambda \min_{i,\alpha} \left| \frac{1}{\tilde{c}_{i,\alpha}|\xi|} \right| \quad (24)$$

where λ is the CFL number ($\lambda \in (0, 1]$). In the computational domain, the dimensionless relaxation times are gained by Eq. (9) as well, but the value of time step is renewed by Eq. (24) locally. The rationality of such treatment will be verified in Section 4.2. Different from that in the standard LB model, the local time step determined by the CFL condition is generally less than one. So then, the dimensionless relaxation time is increased as it is the function of viscosity and local time step. In this aspect, the numerical stability of present model is improved, which can obtain the similar effect by using central moments or multiple relaxation times as those in the work of Hajabdollahi et al. [41,42]. Until now, the 3D thermal LB model on non-orthogonal grid with second-order accuracy is successfully established, and the basic numerical procedures of present model can refer to the Refs. [24,29] for more information.

3. Boundary treatment method

In this paper, the non-equilibrium extrapolation scheme proposed by Guo et al. [37,43] is developed to obtain the unknown density and temperature distribution functions on the irregular boundaries. The basic idea of the extrapolation scheme is that the distribution function is assumed to be composed of its equilib-

rium and non-equilibrium parts. Hence, in the physical domain, the post-collision particle distribution functions at the boundary node $\mathbf{x}_b$ are expressed as

$$\begin{cases} f_i^*(\mathbf{x}_b, t) = f_i^{eq}(\rho(\mathbf{x}_f), \mathbf{u}(\mathbf{x}_b), t) + \left(1 - \frac{1}{\tau_f}\right) [f_i(\mathbf{x}_f, t) - f_i^{eq}(\mathbf{x}_f, t)] \\ h_i^*(\mathbf{x}_b, t) = h_i^{eq}(T(\mathbf{x}_b), \mathbf{u}(\mathbf{x}_b), t) + \left(1 - \frac{1}{\tau_T}\right) [h_i(\mathbf{x}_f, t) - h_i^{eq}(\mathbf{x}_f, t)] \end{cases} \quad (25)$$

where $\mathbf{x}_f$ is the fluid node next to the boundary node $\mathbf{x}_b$ in the physical domain.

Since the governing equations in the physical domain have been transformed into the computational domain, the transformation of Eq. (25) for boundary implementation is stated as follows:

Firstly, through the Chapman-Enskog analysis, the non-equilibrium terms in Eq. (25) can be assumed to be $f_i^{(ne)} = f_i - f_i^{eq} = K f_i^{(1)}$ and $h_i^{(ne)} = h_i - h_i^{eq} = K h_i^{(1)}$ with K being a small parameter proportion to the Knudsen number. Therefore, in the computational domain, the non-equilibrium terms in Eq. (25) can be directly obtained by replacing $\mathbf{x}$ with ξ as

$$\begin{cases} f_i^{(ne)} = f_i - f_i^{eq} = f_i(\xi_f, t) - f_i^{eq}(\xi_f, t) \\ h_i^{(ne)} = h_i - h_i^{eq} = h_i(\xi_f, t) - h_i^{eq}(\xi_f, t) \end{cases} \quad (26)$$

where ξ_f is the fluid node next to the boundary node ξ_b in the computational domain.

Then, special treatment comes from the transformation of equilibrium parts. For Dirichlet type boundary conditions, the equilibrium terms $f_i^{(eq)} = f_i^{eq}$ and $h_i^{(eq)} = h_i^{eq}$ in Eq. (25) can also be determined by replacing $\mathbf{x}$ with ξ as

$$\begin{cases} f_i^{(eq)} = f_i^{eq} = f_i^{eq}(\rho(\xi_f), \mathbf{u}(\xi_b), t) \\ h_i^{(eq)} = h_i^{eq} = h_i^{eq}(T(\xi_b), \mathbf{u}(\xi_b), t) \end{cases} \quad (27)$$

However, it should be emphasized that as for the symmetry boundary or adiabatic boundary which contains the derivative term, the macroscopic variables at boundary node ξ_b in Eq. (27) are unknown. Thus, the following procedure is added to gain their values.

For the sake of generalization, all types of boundary conditions are contained in the following equation,

$$b_1 \phi + b_2 \frac{\partial \phi}{\partial n} = b_3 \quad (28)$$

where ϕ is a generic variable, b_1 , b_2 , and b_3 are constant quantities, $\partial \phi / \partial n$ is the normal derivative of ϕ on the boundary. During the transformation, the constant quantities remain the same while $\partial \phi / \partial n$ is not. To derive the conversion equation of $\partial \phi / \partial n$, we define a function $f(x, y, z) = \text{const}$ with its gradient calculated as

$$\nabla f = \frac{\partial f}{\partial x} \vec{i} + \frac{\partial f}{\partial y} \vec{j} + \frac{\partial f}{\partial z} \vec{k} \quad (29)$$

where $(\vec{i}, \vec{j}, \vec{k})$ are the unit vectors of coordinate axis (x, y, z) . By using the chain rule, the Eq. (29) can be related to computational domain as

$$\nabla f = \frac{\partial f}{\partial \xi_x} \frac{\partial \xi_x}{\partial x} \vec{i} + \frac{\partial f}{\partial \xi_y} \frac{\partial \xi_y}{\partial y} \vec{j} + \frac{\partial f}{\partial \xi_z} \frac{\partial \xi_z}{\partial z} \vec{k} \quad (30)$$

Let $f = \xi$, the unit vector $\vec{n}^{(\xi)}$ perpendicular to $\xi(x, y, z) = \text{const}$ is obtained as

$$\vec{n}^{(\xi)} = \frac{\nabla \xi}{|\nabla \xi|} = \left(\xi_x \vec{i} + \xi_y \vec{j} + \xi_z \vec{k} \right) / \sqrt{\xi_x^2 + \xi_y^2 + \xi_z^2} \quad (31)$$

Similarly, the unit vectors $\vec{n}^{(\eta)}$ and $\vec{n}^{(\gamma)}$ are given as

$$\vec{n}^{(\eta)} = \frac{\nabla \eta}{|\nabla \eta|} = \left(\eta_x \vec{i} + \eta_y \vec{j} + \eta_z \vec{k} \right) / \sqrt{\eta_x^2 + \eta_y^2 + \eta_z^2} \quad (32)$$

$$\vec{n}^{(\gamma)} = \frac{\nabla \gamma}{|\nabla \gamma|} = \left(\gamma_x \vec{i} + \gamma_y \vec{j} + \gamma_z \vec{k} \right) / \sqrt{\gamma_x^2 + \gamma_y^2 + \gamma_z^2} \quad (33)$$

Based on Eqs. (31)–(33), the conversion equation of normal derivative on boundaries with arbitrary shapes can be derived as

$$\frac{\partial \phi}{\partial n^{(\xi)}} = (mm\phi_\xi + mn\phi_\eta + ml\phi_\gamma) / \sqrt{mm} \quad (34)$$

$$\frac{\partial \phi}{\partial n^{(\eta)}} = (mn\phi_\xi + nn\phi_\eta + nl\phi_\gamma) / \sqrt{nn} \quad (35)$$

$$\frac{\partial \phi}{\partial n^{(\gamma)}} = (ml\phi_\xi + nl\phi_\eta + ll\phi_\gamma) / \sqrt{ll} \quad (36)$$

with

$$\begin{cases} mm = \xi_x^2 + \xi_y^2 + \xi_z^2 \\ mn = \xi_x\eta_x + \xi_y\eta_y + \xi_z\eta_z \\ ml = \xi_x\gamma_x + \xi_y\gamma_y + \xi_z\gamma_z \\ nn = \eta_x^2 + \eta_y^2 + \eta_z^2 \\ nl = \eta_x\gamma_x + \eta_y\gamma_y + \eta_z\gamma_z \\ ll = \gamma_x^2 + \gamma_y^2 + \gamma_z^2 \end{cases} \quad (37)$$

The spatial derivatives in Eqs. (34)–(36) are calculated by second-order central-difference scheme or second-order one-sided finite difference scheme, while those in Eq. (37) have been obtained by Eq. (14) in Section 2.2. Then, the unknown macroscopic variables at boundary node ξ_b in Eq. (27) are determined by substituting Eqs. (32)–(34) into Eq. (28). Taking the adiabatic boundary for example, the temperature at boundary node $T(\mathbf{x}_b)$ in the equilibrium term $h_i^{eq}(T(\mathbf{x}_b), \mathbf{u}(\mathbf{x}_b), t)$ is unknown in the physical domain. Instead, the normal derivative at the boundary can be expressed as

$$\frac{\partial T}{\partial n} = 0 \quad (38)$$

In the computational domain, it can be derived as

$$\frac{\partial T}{\partial n^{(\xi)}} = \left(mm \frac{\partial T}{\partial \xi} + mn \frac{\partial T}{\partial \eta} + ml \frac{\partial T}{\partial \gamma} \right) / \sqrt{mm} = 0 \quad (39a)$$

$$\frac{\partial T}{\partial n^{(\eta)}} = \left(mn \frac{\partial T}{\partial \xi} + nn \frac{\partial T}{\partial \eta} + nl \frac{\partial T}{\partial \gamma} \right) / \sqrt{nn} = 0 \quad (39b)$$

$$\frac{\partial T}{\partial n^{(\gamma)}} = \left(ml \frac{\partial T}{\partial \xi} + nl \frac{\partial T}{\partial \eta} + ll \frac{\partial T}{\partial \gamma} \right) / \sqrt{ll} = 0 \quad (39c)$$

Consequently, the unknown temperature at boundary node ξ_b is acquired by solving Eq. (39). The similar manner can be applied to derive the symmetry boundary with $\partial u / \partial n = 0$. By replacing the equilibrium and non-equilibrium parts in Eq. (25) with those in Eq. (26) and Eq. (27) respectively, the transformation of Eq. (25) into the computational domain is realized.

Assisted with above conversion equations, a general method for boundary treatment on non-orthogonal grid is established, by which the adiabatic boundary condition or symmetry boundary condition could be easily implemented on the irregular boundary. Compared with that in the original work of the GILBM used for Dirichlet boundary condition [24], the current boundary treatment method can cover more types of boundary conditions.

4. Numerical results and discussion

4.1. Natural convection in a 3D horizontal concentric air-filled annulus

As shown in Fig. 4, natural convection in a 3D horizontal concentric air-filled annulus is simulated for numerical validation primarily. The annulus with length L is formed by two concentric cylinders of radii r_i (inner) and r_o (outer), holding uniform temperatures $T_i = 1.0$ and $T_o = 0.0$, respectively. Right and left sides of the annulus are bounded by two impermeable and adiabatic plates. Meanwhile, non-slip boundary conditions for the velocity are held at all the walls. Natural convection in the annulus is characterized by four dimensionless parameters: Prandtl number Pr , Rayleigh number Ra , radius ratio $RR = r_o/r_i$, and axial aspect ratio $A = L/(r_o - r_i)$. Due to the symmetry of the 3D annulus, only its half part is considered with symmetric condition applied at $\theta = 0$ and $\theta = \pi$, respectively. The grid generation in the physical and computational domains is illustrated in Fig. 5. The grid is uniform in the computational domain, while a polar coordinate grid is used in the physical domain.

According to the investigation of Powe et al. [30], the flow structure in a 3D annulus is mainly two-dimensional when the radius ratio $RR > 1.71$, while counter-rotating transverse rolls attributed to thermal instability are reported to appear in the upper region of the annulus with $1.24 \leq RR \leq 1.71$. To verify the feasibility of the dimensionless relaxation time based on the local time step, pure conduction in the annulus with $RR = 2.6$, $A = 14.0$ is solved preliminarily. Temperature distribution between the inner and outer cylinders is compared with the theoretical one given as $T = T_i + \ln(r/r_i)(T_o - T_i)/\ln(r_o/r_i)$ in Fig. 6(a), and the maximum deviation is less than 1.05%. When natural convection occurs in the annulus with $Ra = 5 \times 10^4$ and $Pr = 0.7$, the local Nusselt number (Nu) distributions along the inner and outer cylinders at the mid-axial plane agree well with the experiment results of Kuehn and Goldstein [44] in Fig. 6(b). The maximum deviation of Nu is found to be less than 4.64%, which occurs at the corner of the computational domain ($\theta = \pi$). The above comparisons indicate that the dimensionless relaxation times are reasonably determined.

To further test the present model and better illustrate the significant change of thermal flow structure with radius ratio, natural convection in a 3D annulus with $RR = 1.6$, $A = 6.667$ is simulated. The Prandtl number is fixed at $Pr = 0.7$, while the Rayleigh number changes from $Ra = 10^3$ to 2.8×10^3 . By means of a grid refinement study, the following results are all based on the grids $52 \times 82 \times 162$. Fig. 7 shows the distributions of velocity vector and isotherm in the upper symmetry section of the annulus. For $Ra = 10^3$, two incompletely formed rotational cells occur next to the endwalls, and the isotherms are disturbed slightly. Once the Rayleigh number exceeds the linear critical value ($Ra_c = 1.9 \times 10^3$) and it is up to $Ra = 2 \times 10^3$, two complete cells appear next to the endwalls rotating in opposite direction. The corresponding isotherms fluctuate obviously in the region adjacent to the endwalls. When the Rayleigh number increases to $Ra = 2.8 \times 10^3$, three pairs of counter-rotating cells arise leading to the S-shaped waves of isotherm in Fig. 7(b). The above-mentioned results match well with those reported by Petrone et al. [45] and less grids are used, which demonstrates that present method can accurately and effectively forecast the evolution of thermal flow structure in the 3D annulus.

In order to better understand the 3D thermal flow structure in the annulus with $RR = 1.6$ and $A = 6.667$, the distributions of streamlines and isothermal surfaces are presented in Fig. 8 for $Ra = 2.8 \times 10^3$. Initially, the spiral motion of counter-rotating cells

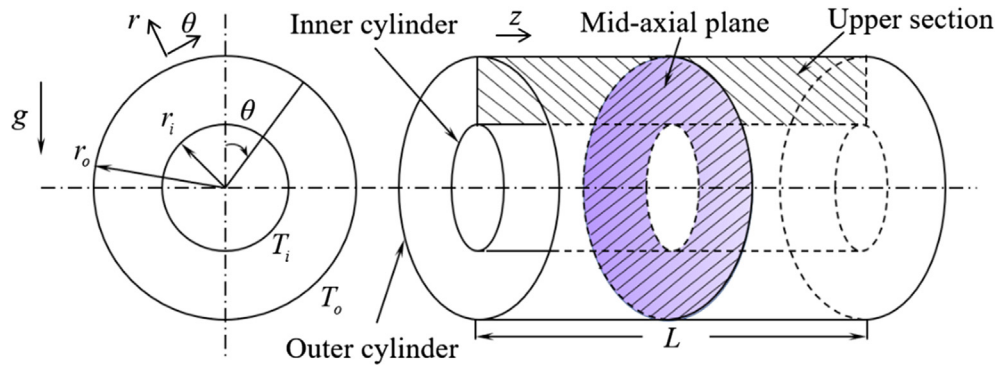


Fig. 4. Schematic diagram of the 3D horizontal concentric annulus.

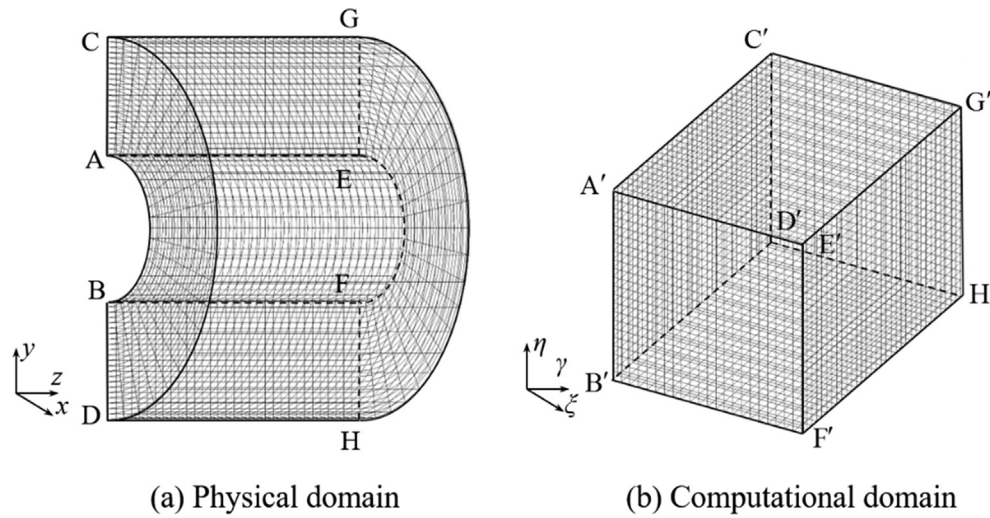


Fig. 5. Grid generation in the (a) physical domain and (b) computational domain of 3D annulus.

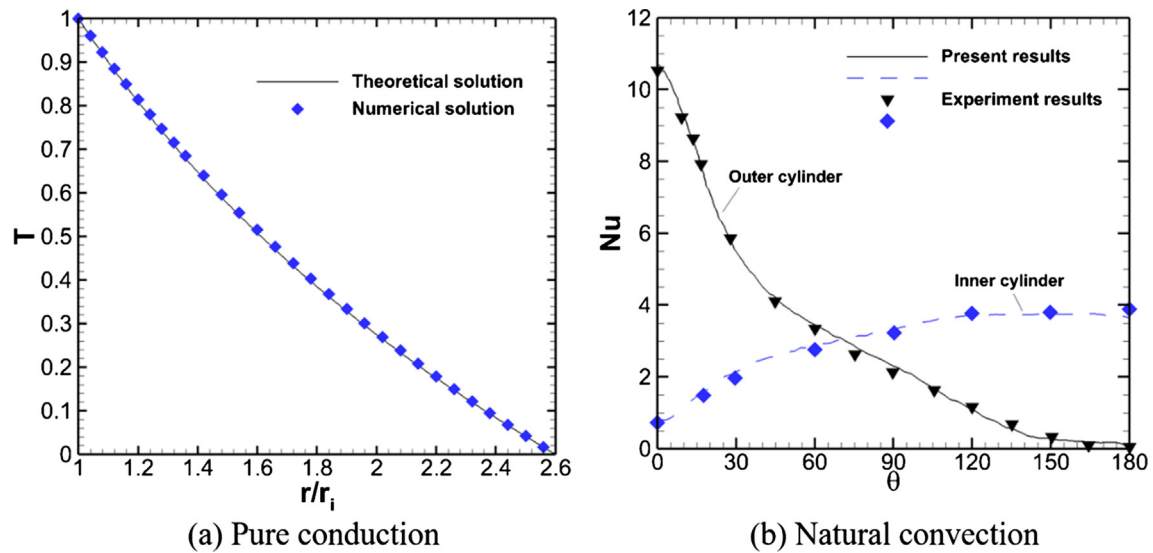


Fig. 6. Results comparison of (a) pure conduction and (b) natural convection in the annulus with $RR = 2.6$, $A = 14.0$.

arises in the upper region of annulus. Then, the streamlines of two end cells are ejected into the core region following a transverse helical motion toward the annulus center, while those of four inner cells are injected into the core region producing a crescent-shaped

flow without a transverse motion. Although the transverse waving isotherms form in the upper region of the annulus, the isotherms in the lower region of the annulus are not perturbed and they keep horizontal along the axial direction. Such a phenomenon is attrib-

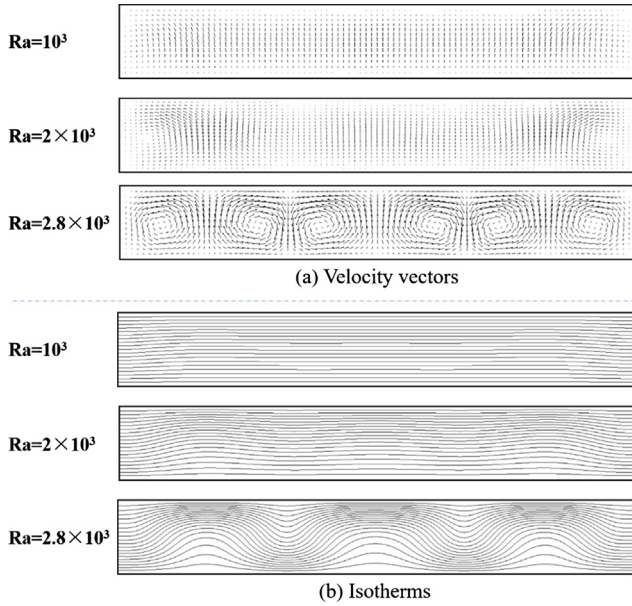


Fig. 7. (a) Velocity vectors and (b) isotherms in the upper section of the annulus with $RR = 1.6$ and $A = 6.667$ for different Rayleigh numbers.

uted to the fact that the thermal disturbance does not reach the bottom region of the annulus with $RR = 1.6$, and it is in line with the observation of Adachi et al. [46].

4.2. Natural convection in a 3D cavity with inclined side walls

Detail of the 3D inclined cavity considered are shown in Fig. 9. The length of all the cavity edges is $H = 1.0$, and non-slip boundary conditions for the velocity are applied for all the walls. Furthermore, the two side walls with inclined angle $\theta = 45^\circ$ are specified with hot ($T_h = 1.5$) and cold ($T_c = 0.5$) temperatures, respectively. The remained four walls are set with two different types of thermal boundary condition. In the Type 1, they are all adiabatic. As for the Type 2, a physically-realizable thermal boundary condition [47] with linear temperature variation from the cold surface to the hot surface is adopted as

$$\begin{cases} y = 0 \text{ and } H, & T = T_h + (T_c - T_h) \cdot x/H \\ z = 0 \text{ and } \sqrt{2}H/2, & T = T_h + (T_c - T_h) \cdot x/H \end{cases} \quad (40)$$

For each of the thermal boundary setting, natural convection in the inclined cavity with two different Prandtl numbers ($Pr = 0.1$ and 10.0) is analyzed at a high Rayleigh number $Ra = 10^6$.

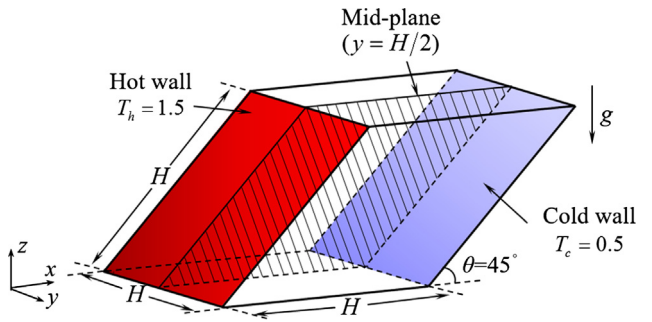
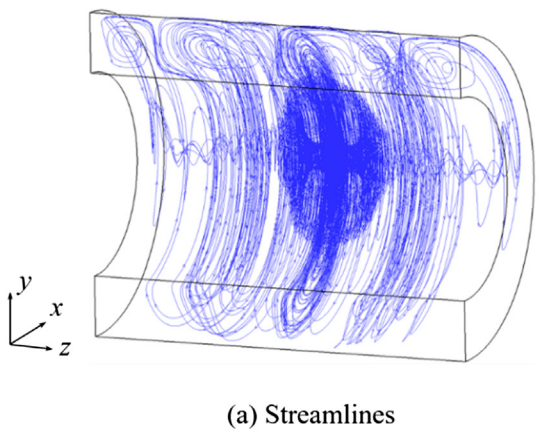


Fig. 9. Schematic diagram of the 3D cavity with inclined side walls.

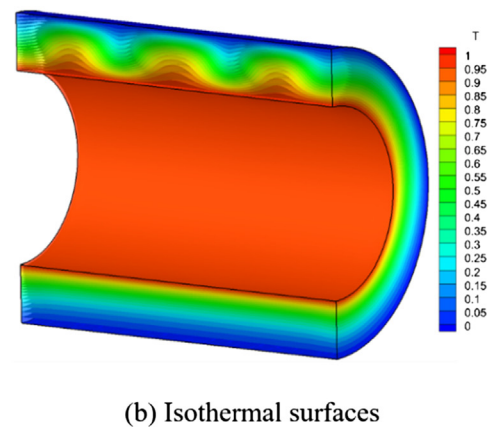


Fig. 8. (a) Streamlines and (b) isothermal surfaces in the annulus with $RR = 1.6$ and $A = 6.667$ for $Ra = 2.8 \times 10^3$.

As the heat transfer near the walls is more intense, a finer grid is used in the near-wall region while a relatively coarse one is adopted in the remaining part. Taking the grid meshing in the x -direction as example, the coordinate component x relates to ξ as

$$x = \begin{cases} a(e^{\xi/b} - 1.0) & , \xi \in [0, NX/2) \\ 1.0 - a(e^{(NX-\xi)/b} - 1.0) & , \xi \in [NX/2, NX] \end{cases} \quad (41)$$

where NX is the grid number in the ξ -direction. a and b are the parameters adjusting the local grid refinement. The grid meshing in the y - and z -directions can be governed similarly. The grid generation in the cavity for both physical and computational domains is then given in Fig. 10.

Grid resolution study is conducted first under Type 1 thermal boundary setting. Table 1 shows the comparison of the average Nusselt number $\bar{Nu}_{H/2}$ and the maximum Nusselt number $Nu_{\max|H/2}$ at cold side of the mid-plane $y = H/2$ between present results and those in Ref. [48]. It can be found out that the results do not change significantly when the grids are increased to 76^3 . Meanwhile, the results obtained by grids 76^3 agree well with the 2D results in the literature. Hence, unless otherwise noted, all the results given in this section are based on the grids 76^3 with local grid refinement factors $a = 0.474$ and $b = 0.015$.

The velocity fields for different Prandtl numbers are portrayed by streamlines in Fig. 11, including 3D results in the whole domain and 2D results on the mid-planes. It is obvious that the Prandtl number has a significant influence on the flow patterns. For $Pr = 0.1$, the central part of the cavity is dominated by a large vortex with two free stagnation points up and down. Yet in the case of $Pr = 10.0$, two secondary vortices emerge which is symmetrical about a free stagnation point located at the cavity center. Another difference is that the main current for $Pr = 0.1$ tends to separate

Table 1
Comparison of $\bar{N}u_{H/2}$ and $Nu_{\max|H/2}$ with those in Ref. [48].

Nu	Pr	Ref. [48]	Present method		
			56^3	76^3	96^3
$\bar{N}u_{H/2}$	0.1	5.985	5.768	5.843	5.875
	10.0	7.580	7.374	7.432	7.459
$Nu_{\max H/2}$	0.1	8.678	8.456	8.551	8.596
	10.0	12.471	12.215	12.362	12.394

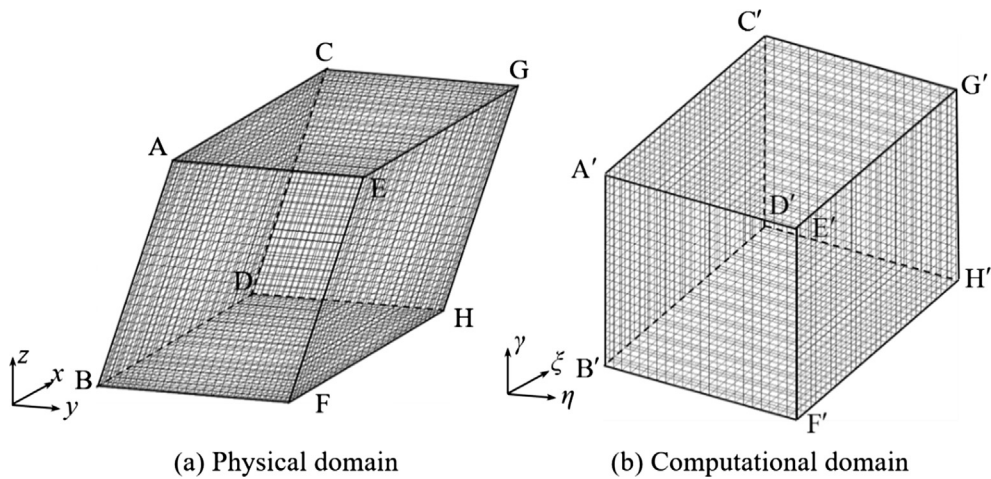


Fig. 10. Grid generation in the (a) physical domain and (b) computational domain of the 3D inclined cavity.

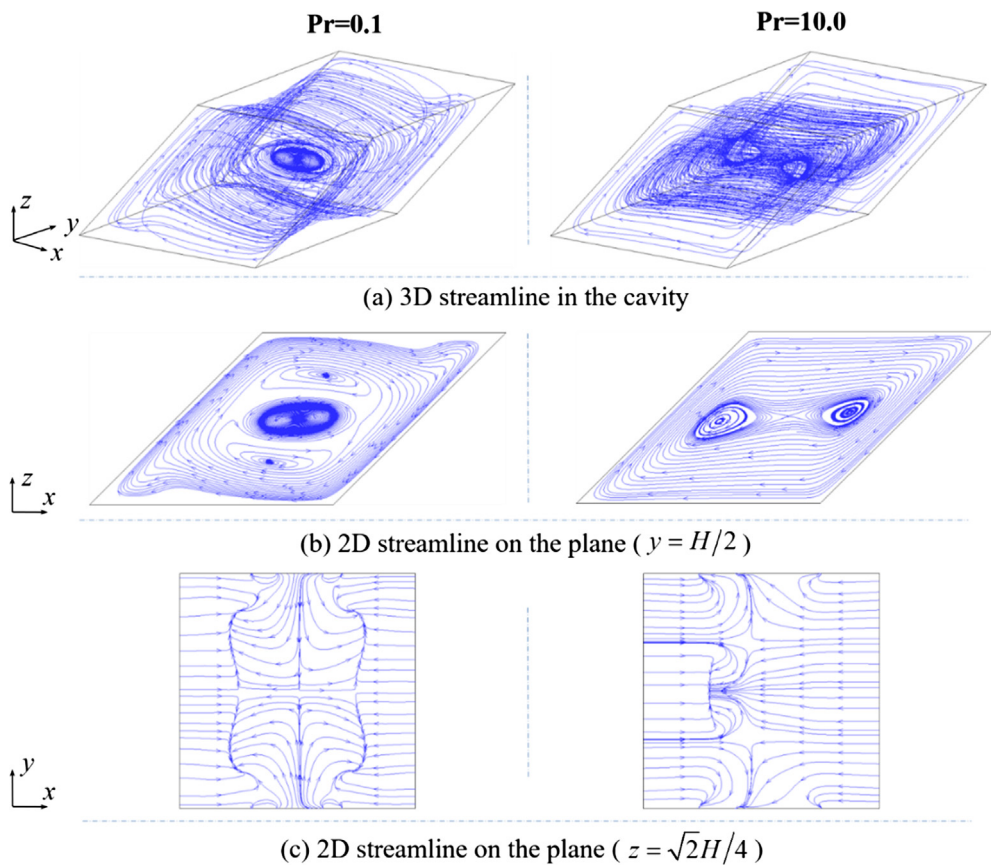


Fig. 11. Streamline distributions with Type 1 thermal boundary setting.

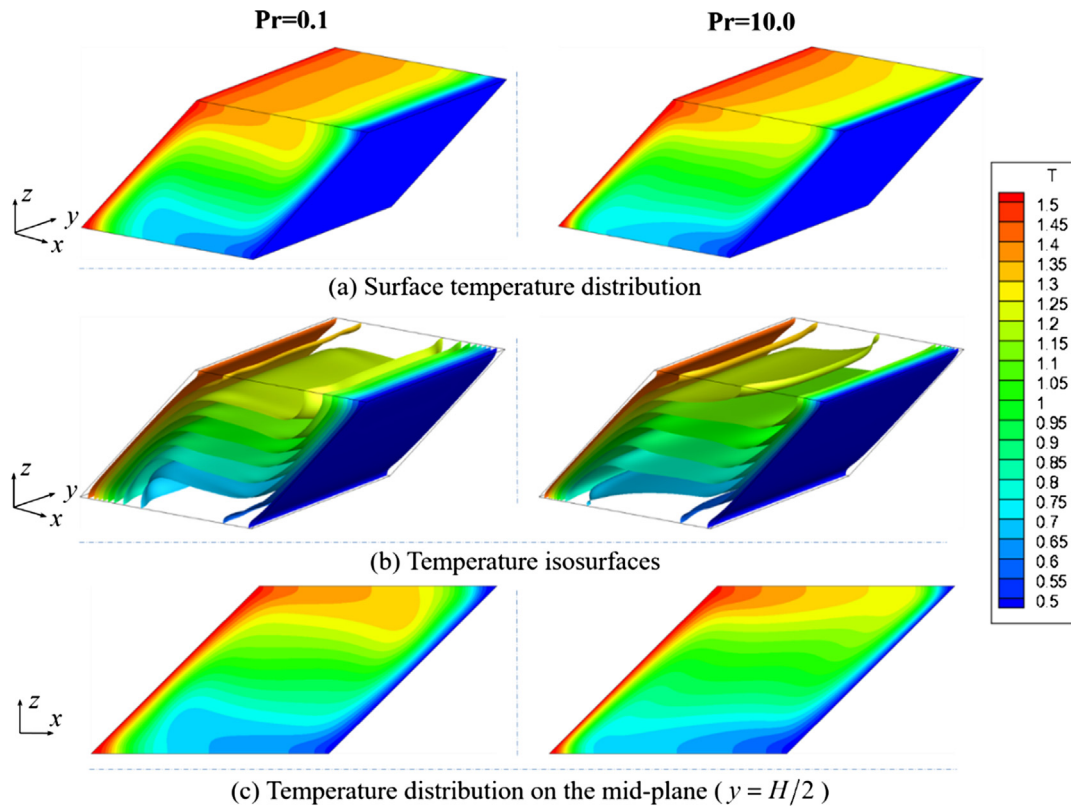


Fig. 12. Temperature distributions with Type 1 thermal boundary setting.

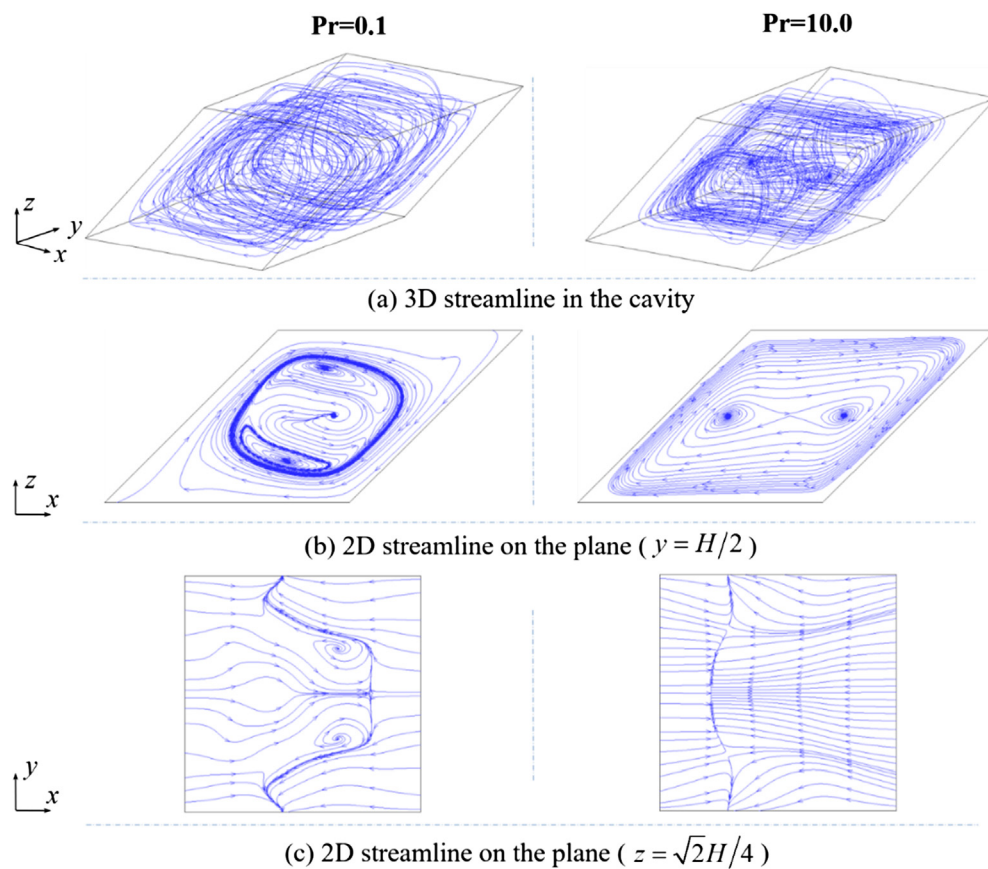


Fig. 13. Streamline distributions with Type 2 thermal boundary setting.

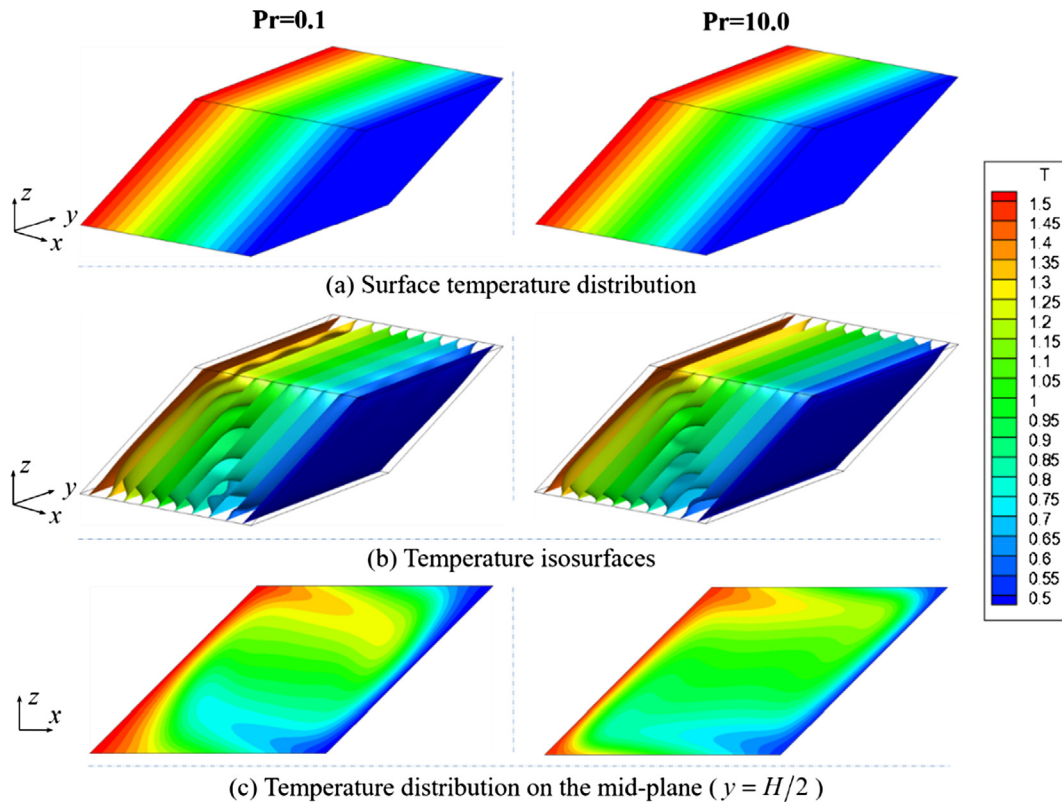


Fig. 14. Temperature distributions with Type 2 thermal boundary setting.

from the upper/lower horizontal surfaces, while the flow keeps attached to the horizontal surfaces for $Pr = 10.0$ due to the enhancement of natural convection at a larger Prandtl number. Meanwhile, as shown in Fig. 11(c), the streamlines on the plane $z = \sqrt{2}H/4$ indicate that fluid flowing to the cavity center in the y -direction joins the two-dimensional flow in the x - and z -directions. Namely, present 3D natural convection can be viewed as a 2D process in the x - and z -directions. It is also proved by that the above-mentioned values of $\bar{Nu}_{H/2}$ and $Nu_{\max|H/2}$ match so well with the 2D results in Ref. [48]. The possible reason is that by setting the two symmetrical walls ($y = 0$ and $y = H$) with the adiabatic boundary condition, the temperature is evenly distributed in the y -direction, which suppresses the fluid flow in this direction. Thus, the fluid tends to flow in the x - and z -directions forming the 2D flow pattern in the 3D inclined cavity. A similar observation is reported by Li et al. [38] when the natural convection occurs in a 3D cubical cavity with Type 1 thermal boundary setting.

Temperature distribution for Type 1 thermal boundary setting are demonstrated in Fig. 12 from different perspectives. Under present high Rayleigh number ($Ra = 10^6$), convection has dominated the heat transfer in the cavity. As shown in Fig. 12(a) and (b), the temperature isosurfaces vary slightly in the y -direction except for the curl near the adiabatic surface ($y = 0$ and H). But, the effect of convection become obvious in the x - and z -directions as illustrated in Fig. 12(c). The isotherms almost paralleling to the isothermal wall turn to be horizontal at the central region, especially for $Pr = 10.0$. Stemmed from the adiabatic boundary condition, the isotherms are orthogonal to the upper and lower surfaces.

Compared to the results of Type 1 thermal boundary setting, distinct 3D features appear under the physically-realizable Type 2 thermal boundary setting. In the streamline profiles, a free stagnation point arises at the central region accompanied by two secondary vortices for $Pr = 0.1$, as shown in Fig. 13(b). Two

intersections form on the plane $z = \sqrt{2}H/4$ breaking the two-dimensional flow pattern in the x - and z -directions. Nevertheless, in the case of $Pr = 10.0$, the streamline distribution own obvious similarity with that of Type 1 condition, indicating that a large Prandtl number can suppress the 3D effect stemmed from thermal boundary setting. Those are because of that at a small Prandtl number ($Pr = 0.1$), the natural convection in the 3D inclined cavity is dominated by the thermal boundary setting. According to the experiment of Leong et al. [47], the uniformity of the temperature field in the y -direction is broken once the thermal boundary setting is changed from Type 1 to Type 2. Then, the distinct temperature gradient enhances the fluid flow in the y -direction and causes the 3D flow pattern in the research domain. Once the Prandtl number is increased to $Pr = 10.0$, the Prandtl number dominates the natural convection rather than the thermal boundary setting. Simultaneously, due to the enhancement of natural convection at a larger Prandtl number, the fluid tends to flow in the x - and z -directions, thus weakening the dimensional effect.

In terms of the temperature distributions given in Fig. 14, isotherms attached to the hot and cold surfaces evolve to be flat in the central region of cavity as the consequence of natural convection. However, the isotherms near the upper and lower surfaces are not orthogonal to them derived from the linear temperature setting. The above-mentioned results demonstrate that both the thermal boundary setting and Prandtl number can significantly influence the hydrodynamic and thermodynamic characteristics of natural convection in the 3D inclined cavity.

5. Concluding remarks

By coupling GILBM with the energy equation, a 3D thermal LB model on non-orthogonal grid is derived in this paper, as well as the corresponding boundary treatment method. Then, it is applied

to compute 3D natural convective heat transfer occurring in irregular domains. Primarily, natural convection in a 3D horizontal concentric air-filled annulus is simulated for numerical validation. The obtained results match well with the those in archival literature or theoretical solution. Afterwards, natural convection in a 3D cavity with inclined side walls is investigated under a high Rayleigh number ($Ra = 10^6$). For different thermal boundary setting, natural convection in the 3D inclined cavity can, generally, be viewed as a 2D process under Type 1 thermal boundary setting, while clear 3D feature appears under Type 2 condition. In terms of Prandtl number, a large Prandtl number ($Pr = 10.0$) suppresses the 3D effect resulted from thermal boundary setting in the cavity. Based on the results above, it can be concluded that present model and boundary treatment method on non-orthogonal grid are reliable in dealing with 3D natural convective heat transfer in irregular domains.

Conflict of interest statement

I would like to declare on behalf of my co-authors that no conflict of interest exists in the submission of this manuscript, and manuscript is approved by all authors for publication. We confirm that the contents of this manuscript is an original research which have not been published previously, and not under consideration for publication elsewhere, in whole or in part.

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